Notes: Shin & Stulz (QJE, 1998)

Are Internal Capital Markets Efficient?

Contents

1 Big picture

- **Question.** Are diversified firms' internal capital markets efficient allocators of capital across segments?
- **Motivation.** A major benefit of diversification is supposed to be that headquarters can use internal capital markets to move funds from cash-rich segments toward segments with better investment opportunities, especially when external finance is costly.
- **Core tension.** Earlier work shows diversified firms often trade at a discount and that diversifying acquisitions can destroy value. Shin and Stulz ask whether this poor performance reflects inefficient internal capital allocation.
- **Empirical setting.** The paper uses Compustat segment data from 1980 to 1992, focusing on the largest and smallest segments of diversified firms.
- **First main fact.** A segment's investment depends significantly on the cash flow of other segments, so internal capital markets are active.
- **Second main fact.** A segment's investment depends much more on its own cash flow than on other segments' cash flow, so internal capital markets are limited.
- **Third main fact.** In highly diversified firms, segment investment is less sensitive to own cash flow than it is for comparable stand-alone firms, consistent with some internal capital reallocation.
- **Efficiency test.** If internal capital markets are efficient, they should protect high-investment-opportunity segments when cash flow is scarce and should reduce investment in lower-opportunity segments when other segments have better opportunities.
- **Main efficiency result.** Segment investment rises with the segment's own industry-q proxy, but it does not respond meaningfully to the investment opportunities of other segments.
- **High-q segment result.** Segments with the best investment opportunities within the firm do not receive special insulation from cash-flow shocks.
- **Interpretation.** Internal capital markets are active but not strongly efficient. They behave partly like a pooling mechanism, but not like a fully disciplined, NPV-ranking allocation system.
- **Takeaway.** The paper documents a form of internal capital market "socialism": segments are affected by firm-wide resource shocks even when their investment opportunities differ.

2 Incremental contribution of the paper

2.1 Contribution relative to diversification discount papers

- Lang and Stulz and Berger and Ofek show that diversified firms are valued below comparable portfolios of focused firms.
- Shin and Stulz move from firm-level valuation to segment-level investment behaviour.
- Instead of only asking whether diversification destroys value, they ask whether the mechanism behind the diversification discount is inefficient capital allocation inside the firm.

Incremental contribution: The paper provides a microeconomic channel for why diversification can fail: internal capital markets do not allocate funds with strong priority to the best investment opportunities.

2.2 Contribution relative to Lamont's oil-shock evidence

- Lamont shows that non-oil divisions of oil firms cut investment after the oil-price shock reduces oil division cash flow.
- That episode proves that internal capital markets transmit shocks across unrelated divisions.
- Shin and Stulz generalize the question to the full Compustat segment universe over 1980–1992.

Incremental contribution: The paper turns a striking industry shock into a broad cross-sectional and panel test of internal capital market activity and efficiency.

2.3 Contribution relative to theoretical models of internal capital markets

- Models such as Stein emphasize efficient headquarters allocating scarce internal funds toward better projects.
- Models such as Scharfstein and Stein or Rajan, Servaes, and Zingales emphasize distorted allocation, rent seeking, and cross-subsidization.
- Shin and Stulz create direct empirical tests that distinguish these views using own cash flow, other cash flow, own q, and other segments' q.

Incremental contribution: The paper shows that diversified firms neither treat segments as fully independent stand-alone firms nor allocate funds as a frictionless efficient internal capital market would. It documents a middle case: active but weakly selective internal capital markets.

2.4 Contribution to empirical corporate finance methods

- The paper uses segment-level capital expenditures, cash flows, sales, and industry-median Tobin's q to construct investment equations.
- It compares smallest and largest segments separately and distinguishes moderately diversified firms from highly diversified firms.
- It benchmarks diversified-firm segments against matched single-segment firms in the same two-digit SIC industry.
- It uses several robustness checks: fixed effects, first differences, sales normalization, narrow sample restrictions, large-firm subsamples, subperiods, and industry subsamples.

Incremental contribution: The empirical design becomes a template for studying capital allocation inside multidivision firms using segment reporting data.

3 Data and measurement

3.1 Segment data source

- The paper uses Compustat Business Information File segment data.
- Segment disclosure is based on SFAS No. 14 and SEC Regulation S-K, which require reporting for segments that represent a sufficiently large share of consolidated operations.
- The sample period is 1980–1992.
- The paper focuses on firms with at least two reported segments and excludes observations with extreme accounting ratios exceeding one.

Interpretation: Segment reporting allows the authors to observe investment and cash flow within diversified firms, rather than treating the diversified firm as a black box.

3.2 Largest and smallest segments

- The paper focuses on the largest and smallest segments of each diversified firm, using segment sales to measure segment size.
- It excludes cases in which the largest and smallest segments have the same two-digit SIC code.
- This ensures that the largest and smallest segments are economically distinct and not simply alternative reporting units within the same industry.
- The final broad sample has 13,670 firm-year records corresponding to 2,631 firms.

Interpretation: The largest-smallest comparison is designed to test whether capital allocation differs between dominant and peripheral divisions.

3.3 Moderate versus high diversification

- Firms are split into two groups: firms with two to four segments and firms with five or more segments.
- This distinction captures the idea that internal capital markets may matter more in highly diversified firms.
- In highly diversified firms, the smallest segment is more likely to be small relative to firm resources, so headquarters should be able to protect it if it has good opportunities.

Interpretation: If internal capital markets work well, the difference between own-segment cash flow and firm-level resources should matter less in highly diversified firms.

3.4 Investment measure

For segment i of firm j in year t, the dependent variable is:

$$I_{ijt} = \frac{CapitalExpenditure_{ijt}}{Assets_{ij,t-1}}. \tag{1}$$

Depending on the specification, the denominator is either segment assets, firm assets, or segment sales.

Interpretation: Normalization is important because segment size varies dramatically across firms and industries.

3.5 Cash flow variables

The two central cash flow variables are:

$$OwnCF_{ijt} = \frac{CashFlow_{ijt}}{Assets_{ij,t-1}}, \tag{2}$$

$$OtherCF_{ijt} = \frac{\sum_{k\neq i} CashFlow_{kjt}}{Assets_{j,t-1}}. \tag{3}$$

- **Own cash flow** measures the segment's internal liquidity.
- **Other cash flow** measures resources generated by the firm's other segments.

Interpretation: If internal capital markets are strong, other segments' cash flow should matter because it measures firm-wide resources. If segments are financially autonomous, own cash flow should dominate.

3.6 Investment opportunities

Investment opportunities are proxied by industry Tobin's q:

$$q_{ijt} = \text{median Tobin's } q \text{ of single-segment firms in segment } i\text{'s industry.} \tag{4}$$

The paper also ranks all segments inside a diversified firm by their industry q values.

Interpretation: A segment's own industry q should predict its investment if the firm pays attention to investment opportunities. Other segments' q should affect its investment if headquarters reallocates scarce funds toward the best opportunities.

4 Empirical specifications

4.1 Baseline segment investment equation

The baseline specification is:

$$I_{ijt} = \alpha_i + \delta_t + \beta_1 SalesGrowth_{ijt} + \beta_2 OwnCF_{ijt} + \beta_3 OtherCF_{ijt} + \beta_4 q_{ij,t-1} + \varepsilon_{ijt}. \tag{5}$$

- α_i captures segment fixed effects.
- δ_t captures year effects.
- $SalesGrowth$ controls for recent demand growth.
- $OwnCF$ captures the segment's own liquidity.
- $OtherCF$ captures resources generated by other divisions.
- q captures the segment's investment opportunities.

Interpretation: The coefficient β_3 tests whether internal capital markets are active. The comparison between β_2 and β_3 tests whether segments rely more on their own cash flow than on other divisions' cash flow.

4.2 Benchmarking against single-segment firms

The paper compares diversified-firm segments with stand-alone firms in the same two-digit SIC industry:

$$I_{ijt} = \alpha_i + \delta_t + \beta_1 SalesGrowth_{ijt} + \beta_2 OwnCF_{ijt} + \beta_3 q_{ij,t-1} + \gamma Single_i + \lambda(Single_i \times OwnCF_{ijt}) + \varepsilon_{ijt}. \tag{6}$$

Interpretation: A positive λ means single-segment firms' investment is more sensitive to own cash flow than comparable diversified segments' investment.

4.3 Robustness specifications

The paper estimates several alternative specifications:

- fixed-effects regressions normalizing by segment assets;
- first-difference regressions;
- ordinary least squares regressions normalizing by segment sales, following Lamont;
- a narrow sample that excludes segments around changes in segment count or SIC code;
- large-firm subsamples;
- subperiod regressions for 1980–1985 and 1986–1992;
- industry subsamples using one-digit SIC codes.

Interpretation: These checks address concerns about accounting noise, changes in segment reporting, scaling, time variation, and industry-specific behaviour.

4.4 Efficiency specification using segment rankings by q

To test whether high-opportunity segments are protected, the paper estimates:

$$I_{ijt} = \alpha_i + \delta_t + \beta_1 SalesGrowth_{ijt} + \beta_2 OwnCF_{ijt} + \beta_3 OtherCF_{ijt} + \beta_4 q_{ij,t-1} \tag{7}$$

$$+ \gamma_1 HighQ_{ijt} \times OwnCF_{ijt} + \gamma_2 HighQ_{ijt} \times OtherCF_{ijt} + \sum_{k \neq i} \rho_k q_{kjt} + \varepsilon_{ijt}. \tag{8}$$

- $HighQ_{ijt}$ equals one if segment i has the highest q among the firm's segments.
- γ_1 and γ_2 test whether the best-opportunity segment is less sensitive to own cash flow and other cash flow.
- ρ_k tests whether other segments' investment opportunities affect the focal segment's investment.

Interpretation: Efficient internal capital markets predict weaker cash-flow sensitivity for high-q segments and negative effects of other segments' q on a given segment's investment.

5 Identification and empirical design

5.1 What identifies internal capital market activity

- The key variation is within diversified firms across segments and over time.
- If one segment's investment responds to other segments' cash flow, the firm's internal capital market is transmitting resources or constraints across divisions.
- Segment fixed effects control for persistent differences in segment investment intensity.
- Year effects absorb common macroeconomic shocks.

Interpretation: The coefficient on other cash flow is the central test of whether divisions are linked through the firm's internal capital market.

5.2 What identifies inefficiency

- Inefficiency is inferred from the weak relationship between capital allocation and relative investment opportunities.
- If headquarters ranks projects efficiently, high-q segments should be protected when cash flow falls.
- If other segments have better q, the focal segment's investment should be lower.
- The paper finds little evidence of these patterns.

Interpretation: The inefficiency test is not simply whether investment depends on cash flow. It asks whether cash-flow dependence varies in the way an efficient internal market would require.

5.3 Why the design is not a full causal experiment

- Segment cash flows are not randomly assigned.
- Industry q is an imperfect proxy for a segment's true investment opportunities.
- Headquarters may observe private information not captured by industry q.
- Segment accounting data may suffer from reporting changes, transfer pricing, and allocation conventions.

Interpretation: The paper's evidence is best read as a strong diagnostic test of internal capital market behaviour, not as a randomized causal estimate.

5.4 Why the evidence is still informative

- Own cash flow and other cash flow have very different coefficients.
- The results survive many normalizations and sample restrictions.
- Other segments' q generally does not explain investment.
- High-q segments are not consistently shielded from cash-flow shocks.

Interpretation: These patterns line up with active but weakly selective internal capital markets.

6 Tables and figures

The paper contains tables rather than plotted figures. Nearby tables are placed on the same row when readability allows.

6.1 Tables I and II: internal capital market activity and stand-alone benchmarks

Read:

- Table I shows that segment investment depends on sales growth, own cash flow, other cash flow, and segment q.
- For the smallest segments in two-to-four segment firms, own cash flow has coefficient 0.1440 while other cash flow has coefficient 0.0182.
- For the largest segments in two-to-four segment firms, own cash flow has coefficient 0.1512 while other cash flow has coefficient 0.0449.
- The own-cash-flow coefficient is always much larger than the other-cash-flow coefficient.
- Table II shows that comparable single-segment firms are more sensitive to own cash flow than segments of diversified firms.

Economic message: Internal capital markets are active because other cash flow matters, but they are limited because own cash flow matters much more.

TABLE I

ESTIMATES OF INVESTMENT EQUATIONS FOR SMALL AND LARGE SEGMENTS

	For the smallest segments only				For the largest segments only			
	2 to 4		5 or more		2 to 4		5 or more	
Sales growth	0.0033	(3.39)	0.0014	(1.61)	0.0098	(3.71)	0.0122	(2.25)
Own cash flow	0.1440	(10.03)	0.1225	(3.39)	0.1512	(14.35)	0.1359	(5.91)
Other cash flow	0.0182	(6.40)	0.0213	(3.41)	0.0449	(4.21)	0.0214	(1.34)
Segment's q	0.0107	(5.38)	0.0053	(4.46)	0.0231	(6.92)	0.0237	(4.12)
Adj-R^2	0.0834		0.0874		0.0863		0.0942	
Obs.	10,437		1886		10,676		2000	

The regressions use all the firms in our sample from 1980 to 1992 for which we can compute the ratios used in our regressions, and we exclude firms where one or more of the accounting ratios used in the regressions exceed one. The dependent variable is the segment's gross capital expenditure for year t divided by the book value of total assets of the firm at the end of year $t-1$. Sales growth is the change in sales from the year $t-2$ to year $t-1$ divided by sales of year $t-2$. Own cash flow is own segment's cash flow for year t divided by the book value of total assets of the firm at the end of year $t-1$. Other cash flow is other segments' cash flow for year t divided by the book value of total assets of the firm at the end of year $t-1$. Segment's q is segment's industry median q at the end of year $t-1$. t-statistics are in parentheses.

Table I: segment investment equations

TABLE II

ESTIMATES OF INVESTMENT EQUATIONS FOR MULTISEGMENT FIRMS AND SINGLE-SEGMENT FIRMS

	For the smallest segments only				For the largest segments only			
	2 to 4		5 or more		2 to 4		5 or more	
Sales growth	0.0410	(13.19)	0.0346	(4.66)	0.0494	(15.51)	0.0446	(5.67)
Own cash flow	0.0757	(11.81)	0.1033	(5.59)	0.1235	(17.99)	0.1380	(9.47)
Segment's q	0.0237	(7.51)	0.0288	(4.04)	0.0193	(7.53)	0.0255	(3.79)
Single-segment firm dummy	−0.00048	(−2.78)	−0.0071	(−1.70)	0.0020	(1.03)	−0.0089	(−1.87)
Single-segment firm dummy*								
Own cash flow	0.0242	(2.64)	0.0683	(2.52)	0.0131	(1.22)	0.1236	(4.27)
Adj-R^2	0.0696		0.0875		0.0931		0.1598	
Obs.	21,792		4202		22,012		4308	

We use all firms in our sample from 1980 to 1992. For each segment we find a stand-alone firm in the same two-digit SIC code that has the asset size closest to the asset size of the segment. These matching firms form our sample of single-segment firms. The single-segment dummy takes the value of one if a firm has only one segment, and zero otherwise. The dependent variable is gross capital expenditures for year t divided by the book value of segment assets at the end of year $t-1$. Sales growth is the change in sales from the year $t-2$ to year $t-1$ divided by sales of year $t-2$. Cash flow is the segment's cash flow for year t divided by the segment's assets at the end of year $t-1$. Segment's q is segment's industry median q at the end of year $t-1$. t-statistics

Table II: diversified segments versus single-segment firms

6.2 Table III: robustness of the investment equations

Read:

- Regression 1 normalizes investment and cash flow by segment assets and allows fixed effects.
- Regression 2 uses first differences.
- Regression 3 follows Lamont's sales-normalized OLS specification.
- Regression 4 uses the narrow sample with more stable segment definitions.
- Across these specifications, the basic finding survives: investment responds to other segments' cash flow, but own cash flow remains important.

Economic message: The internal capital market result is not an artifact of one scaling choice or one sample definition.

TABLE III

OTHER REGRESSION ESTIMATES FOR SMALL AND LARGE SEGMENTS

Panel A. Firms with two to four segments

	For the smallest segments only							
	Regression 1		Regression 2		Regression 3		Regression 4	
Sales growth	0.0033	(0.84)	0.0003	(1.81)	0.0214	(3.82)	0.0036	(3.31)
Own cash flow	0.0926	(10.54)	0.0889	(4.06)	0.1857	(15.84)	0.1220	(6.19)
Other cash flow	0.0488	(5.69)	0.0195	(3.22)	0.1215	(8.94)	0.0227	(6.18)
Segment's q	0.0364	(5.78)	0.0108	(3.83)	0.0266	(4.55)	0.0126	(4.91)
Adj-R^2	0.0491		0.0228		0.1124		0.0753	
Obs.	10,437		10,473		10,757		7316	

	For the largest segments only							
	Regression 1		Regression 2		Regression 3		Regression 4	
Sales growth	0.0135	(3.03)	0.0038	(1.96)	0.0132	(2.82)	0.0088	(2.88)
Own cash flow	0.1384	(15.18)	0.0669	(2.37)	0.3313	(23.39)	0.1485	(12.24)
Other cash flow	0.0247	(3.45)	0.0553	(4.39)	0.0311	(5.44)	0.0473	(3.79)
Segment's q	0.0367	(7.04)	0.0229	(4.70)	0.0132	(3.14)	0.0214	(6.14)
Adj-R^2	0.0840		0.0264		0.2298		0.0826	
Obs.	10,676		10,864		11,170		7491	

Panel B. Firms with five or more segments

	For the smallest segments only							
	Regression 1		Regression 2		Regression 3		Regression 4	
Sales growth	0.0052	(0.60)	−0.0003	(−1.47)	0.0140	(0.82)	0.0027	(2.40)
Own cash flow	0.0604	(2.75)	0.1849	(1.93)	0.1351	(6.02)	0.1400	(4.17)
Other cash flow	0.1060	(3.31)	0.0180	(1.70)	0.2021	(4.38)	0.0316	(3.08)
Segment's q	0.0628	(5.14)	−0.0018	(−0.57)	0.0270	(1.87)	0.0082	(4.38)
Adj-R^2	0.0561		0.0411		0.0708		0.1229	
Obs.	1886		1857		2035		939	

	For the largest segments only							
	Regression 1		Regression 2		Regression 3		Regression 4	
Sales growth	0.0287	(2.73)	−0.0020	(−2.11)	0.0204	(1.72)	0.0121	(1.92)
Own cash flow	0.1108	(5.54)	0.0567	(1.71)	0.3882	(11.41)	0.1213	(5.10)
Other cash flow	0.0352	(1.62)	0.0393	(1.99)	−0.0123	(−0.62)	0.0203	(0.92)
Segment's q	0.0441	(4.24)	0.0156	(2.28)	0.0082	(0.83)	0.0240	(3.17)
Adj-R^2	0.0796		0.0208		0.2460		0.0776	
Obs.	2000		2008		2209		1058	

Regression 1 shows fixed-effects estimates normalizing investment and cash flow by segment assets. Regression 2 shows estimates of a regression in first differences. Regression 3 shows regression of investment and cash flow normalized by segment sales. Regression 4 shows fixed-effect estimates for a narrow sample. t-statistics are in parentheses.

6.3 Table IV: do high-q segments receive priority?

Read:

- Table IV interacts cash flow with a dummy for whether the segment has the highest q inside the firm.
- The interaction terms for high-q status and other cash flow are economically small and statistically insignificant in most cases.
- The high-q interaction with own cash flow is significantly negative only for the smallest segment in firms with two to four segments.
- Other segments' q values are generally insignificant, except for the largest segment in firms with two to four segments.

Economic message: The best-opportunity segment is not consistently protected from cash-flow shocks. The internal capital market does not allocate funds with strong priority to high-q segments.

TABLE IV

ESTIMATES OF INVESTMENT EQUATIONS ALLOWING FOR A DIFFERENT COEFFICIENT FOR THE CASH FLOW OF OTHER SEGMENTS FOR SEGMENTS WITH THE HIGHEST q IN THE FIRM AND USING THE q OF THE OTHER SEGMENTS AS EXPLANATORY VARIABLES

	For the smallest segments only				For the largest segments only			
	2 to 4		5 or more		2 to 4		5 or more	
Sales growth	0.0033	(3.41)	0.0014	(1.60)	0.0093	(3.51)	0.0119	(2.24)
Own cash flow	0.1668	(9.24)	0.1320	(3.36)	0.1508	(13.11)	0.1442	(6.58)
Other cash flow	0.0167	(5.36)	0.0211	(3.16)	0.0468	(4.01)	0.0200	(1.60)
Segment's q	0.0116	(5.32)	0.0055	(4.54)	0.0223	(5.96)	0.0252	(4.09)
Dummy for segment's q* own cash flow	−0.0452	(−2.29)	−0.0371	(−0.79)	0.0026	(0.22)	−0.0225	(−0.65)
Dummy for segment's q* other cash flow	0.0041	(1.10)	−0.0009	(−0.16)	−0.0173	(−1.19)	0.0025	(0.07)
Other segment 1's q	−0.0015	(−1.02)	0.0009	(0.33)	0.0057	(1.26)	−0.0011	(−0.10)
Other segment 2's q	−0.0004	(−0.40)	−0.0013	(−0.57)	0.0056	(3.41)	0.0003	(0.06)
Other segment 3's q	−0.0010	(−1.89)	0.0004	(0.32)	−0.0007	(−0.64)	−0.0046	(−1.40)
Other segment 4's q			0.0008	(1.44)			0.0023	(1.02)
Other segment 5's q			−0.0003	(−0.34)			0.0001	(0.07)
Other segment 6's q			0.0000	(0.06)			−0.0019	(−1.70)
Other segment 7's q			−0.0016	(−1.98)			−0.0002	(−0.15)
Adj-R^2	0.0854		0.0878		0.0881		0.0932	
Obs.	10,437		1886		10,676		2000	
F-value	1.4111		1.0127		7.2971		0.8046	
Prob > F	0.2374		0.4201		0.0001		0.5834	

Regressions use all the firms in our sample from 1980 to 1992 for which we can compute the ratios used in our regressions and we exclude firms where one or more of the accounting ratios used in the regressions exceed one. Dummy for segment's q takes the value of one if segment's q is the highest q in the firm for that year; it equals zero otherwise. F-value is for the joint hypothesis that coefficients of other segments' q are zero. t-statistics are reported in parentheses.

6.4 Tables V and VI: sample composition and descriptive statistics

Read:

- Table V reports the distribution of firm-year observations across years and number of segments.
- The sample has 13,670 firm-year records and 2,631 different firms.
- The number of diversified firm observations declines over time, consistent with declining diversification during the 1980s.
- Table VI shows that other segments' cash flow is extremely large relative to own cash flow for small segments.
- For the smallest segment in firms with two to four segments, mean own cash flow is 24.90 million dollars and mean

other cash flow is 181.67 million dollars.

- For the smallest segment in firms with five or more segments, mean own cash flow is 38.87 million dollars and mean other cash flow is 502.06 million dollars.

Economic message: The descriptive statistics show that diversified firms often have enough firm-wide resources to cushion small segments, yet investment remains highly sensitive to segment-specific cash flow.

TABLE V
Distribution of the Sample Firms across Years and across Number of Segments

Frequency	2	3	4	5	6	7	8	9	10	Total
	Number of segments									
80	391	410	288	156	74	22	14	4	7	1366
81	380	380	282	144	63	26	15	5	3	1298
82	386	365	268	139	60	25	10	4	3	1260
83	384	351	266	126	55	23	11	1	4	1221
84	348	353	242	110	38	19	9	4	3	1126
85	320	327	212	91	32	17	6	5	2	1012
86	360	313	196	79	34	18	6	3	1	1010
87	317	286	195	76	30	12	7	3	2	928
88	320	282	174	75	31	18	4	2	3	909
89	301	285	170	64	33	17	6	1	4	881
90	326	297	177	69	41	16	5	1	4	936
91	350	297	196	78	29	19	2	3	4	978
92	248	238	157	61	26	7	3	2	3	745
Total	4431	4184	2823	1268	546	239	98	38	43	13670

All data come from the Business Information file of COMPUSTAT. We use all firms from 1980 to 1992 for

Table V: sample distribution

TABLE VI
Descriptive Statistics for Initial Sample Data Set

	Firms with 2 to 4 segments							
	For the smallest segments only				For the largest segments only			
	Gross capital expenditure	Sales	Own cash flow	Other cash flow	Gross capital expenditure	Sales	Own cash flow	Other cash flow
Mean	13.93	177.55	24.90	181.67	58.66	968.06	115.81	94.15
99%	249.08	2547.60	379.50	2901.00	870.55	12976.00	1724.00	1487.75
95%	60.80	803.20	113.90	726.00	237.60	3810.40	499.67	332.00
90%	27.01	407.20	57.30	366.97	115.40	1981.00	250.20	159.87
Median	0.93	25.00	2.49	18.09	4.49	129.90	13.74	5.98
10%	0.00	1.41	−0.37	0.48	0.11	9.69	0.32	−0.08
5%	0.00	0.46	−2.02	−0.25	0.03	4.66	−0.43	−1.10
1%	0.00	0.03	−21.00	−11.24	0.00	0.85	−12.59	−16.17

	Firms with 5 or more segments							
	For the smallest segments only				For the largest segments only			
	Gross capital expenditure	Sales	Own cash flow	Other cash flow	Gross capital expenditure	Sales	Own cash flow	Other cash flow
Mean	25.41	235.07	38.87	502.06	98.67	1735.37	180.98	359.94
99%	391.60	2853.00	688.43	5655.00	1560.54	21232.00	2472.61	4864.00
95%	107.00	977.00	164.80	2014.00	362.02	7500.00	711.00	1555.00
90%	53.00	554.00	87.63	1271.00	239.00	4229.30	434.98	923.30
Median	3.03	66.73	7.35	153.90	14.80	510.90	51.81	92.83
10%	0.00	2.05	−1.41	5.11	0.24	38.10	1.46	2.12
5%	0.00	0.60	−7.86	1.02	0.00	16.00	−0.70	−0.64
1%	0.00	0.01	−60.12	−38.22	0.00	4.69	−125.00	−53.75

Table VI: descriptive statistics

7 Short summary

- Shin and Stulz study whether internal capital markets allocate funds efficiently across segments of diversified firms.
- Segment investment depends on other segments' cash flow, so internal capital markets are active.
- Segment investment depends much more on its own cash flow, so internal capital markets are limited.
- Diversified segments are less cash-flow sensitive than stand-alone firms in some settings, especially highly diversified firms.
- Segment investment rises with own industry q, meaning investment opportunities matter.
- Other segments' q generally does not affect a segment's investment.
- High-q segments are not consistently protected from own or other cash-flow shocks.
- The results are robust to fixed effects, first differences, alternative normalizations, narrow samples, large-firm samples, subperiods, and industry subsamples.
- The evidence supports the view that internal capital markets are active but not strongly efficient.
- The results are consistent with internal capital market "socialism" or weakly selective cross-subsidization rather than clean NPV-based capital budgeting.

8 Conclusion

8.1 Core conclusion

Shin and Stulz conclude that internal capital markets in diversified firms are active but not efficient in the strong sense predicted by theories of value-creating diversification. Segment investment responds to firm-wide cash flows, but it responds much more strongly to the segment's own cash flow. Moreover, segments with better investment opportunities are not reliably insulated from cash-flow shocks.

8.2 Economic intuition

An efficient internal capital market should allocate scarce funds to the highest-NPV opportunities. This means that a high-q segment should be less constrained by its own cash flow and less affected by cash flow shocks elsewhere in the firm. The evidence does not support this benchmark. Instead, all segments appear exposed to cash-flow shocks in ways that do not strongly reflect their relative opportunities.

8.3 Takeaway

The paper's lasting message is that diversification does not automatically create an efficient internal capital market. Headquarters can move resources across divisions, but it does not necessarily direct them to the best opportunities. This helps explain why diversified firms can trade at a discount even when internal capital markets are active.